# Data Science Consulting - Heart Disease Analytics

By Utsav Nimavat, Edward Lin, Tiexuan Zhu, Selim Gurkas

# Company Background

Our company, CardioInsight Analytics, is focused on identifying patterns and relationships from patient heart data.

# Initial Data analysis findings

- Our dataset, `heart_disease_uci.csv`, focuses on a mix of heart & chest–pain related attributes for patients and the severity of heart disease they suffer from.

- Categorical attributes include:
  - `sex` (male or female)
  - `dataset` (where the data is from)
  - `cp` (type of chest pain)
  - `fbs` (high level of blood sugar)
  - `restecg` (resting ECG results)
  - `exang` (exercise-induced chest pain)
  - `slope` (trend in stress ECG)
  - `thal` (result of thallium Stress Test)

- Numerical attributes include:
  - `age` (age of patient)
  - `trestbps` (resting blood pressure)
  - `chol` (cholesterol level in patient)
  - `thalch` (maximum heart rate)
  - `oldpeak` (a drop in an ECG line induced by exercise relative to rest. measures severity of oxygen deprivation)
  - `ca` (# of fluoroscopy-affected vessels)
  - `num` (predicted level of heart disease)

| | id | age | sex | dataset | cp | trestbps | chol | fbs | restecg | thalch | exang | oldpeak | slope | ca | thal | num |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 63 | Male | Cleveland | typical angina | 145.0 | 233.0 | True | lv hypertrophy | 150.0 | False | 2.3 | downsloping | 0.0 | fixed defect | 0 |
| **1** | 2 | 67 | Male | Cleveland | asymptomatic | 160.0 | 286.0 | False | lv hypertrophy | 108.0 | True | 1.5 | flat | 3.0 | normal | 2 |
| **2** | 3 | 67 | Male | Cleveland | asymptomatic | 120.0 | 229.0 | False | lv hypertrophy | 129.0 | True | 2.6 | flat | 2.0 | reversable defect | 1 |
| **3** | 4 | 37 | Male | Cleveland | non-anginal | 130.0 | 250.0 | False | normal | 187.0 | False | 3.5 | downsloping | 0.0 | normal | 0 |
| **4** | 5 | 41 | Female | Cleveland | atypical angina | 130.0 | 204.0 | False | lv hypertrophy | 172.0 | False | 1.4 | upsloping | 0.0 | normal | 0 |

# Data Quality

**Row**(observations):

920 patients in our dataset

**Columns**(features):

16

**Memory Usage:**

0.4MB

**Duplicate Rows:**

0

**Missing Values:**

| | Missing Count | Missing Percent (%) |
|---|---|---|
| ca | 611 | 66.41 |
| thal | 486 | 52.83 |
| slope | 309 | 33.59 |
| fbs | 90 | 9.78 |
| oldpeak | 62 | 6.74 |
| trestbps | 59 | 6.41 |
| thalch | 55 | 5.98 |
| exang | 55 | 5.98 |
| chol | 30 | 3.26 |
| restecg | 2 | 0.22 |

# Clinical Focus - Heart Disease by Age

- age_group, was created grouping patients via age:
  - 0-40 (low risk)
  - 40-50 (risk begins accumulating)
  - 50-60 (moderate risk)
  - 60-70 (high risk)
  - 70+ (highest risk)
  - Aggregating age groups over our numeric cols reveal that older patients had higher blood pressure and cholesterol levels, and lower maximum heart rates.
- heart_disease was created, classifying patients if they had heart disease or not.
  - Surprisingly, those from the higher risk period had a highest proportion of heart disease than the highest.

```
result = df.groupby('age_group').agg(
    total_patients=('id', 'count'),
    avg_trestbps= ('trestbps', 'mean'),
    avg_chol = ('chol', 'mean'),
    avg_thalch = ('thalch', 'mean'),
    avg_oldpeak = ('oldpeak', 'mean')
).round(1)
result
```

| age_group | total_patients | avg_trestbps | avg_chol | avg_thalch | avg_oldpeak |
|---|---|---|---|---|---|
| Low baseline cardiovascular risk | 93 | 124.6 | 205.9 | 156.3 | 0.4 |
| Risk factors begin accumulating | 224 | 128.0 | 216.1 | 145.0 | 0.6 |
| Moderate risk period | 382 | 133.5 | 197.8 | 134.7 | 0.9 |
| Higher risk period | 197 | 137.7 | 180.5 | 126.2 | 1.3 |
| Highest risk population | 24 | 139.0 | 194.7 | 121.0 | 1.2 |

```
df['heart_disease'] = np.where(df['num'] >= 1, 1, 0)
result = df.groupby('age_group').agg(
    total_patients=('id', 'count'),
    total_with_disease=('heart_disease', 'sum'),
    percentage = ('heart_disease', 'mean'))
result['percentage'] = ((result['percentage'] * 100).round(1)).astype(str) + '%'
result
```

| age_group | total_patients | total_with_disease | percentage |
|---|---|---|---|
| Low baseline cardiovascular risk | 93 | 32 | 34.4% |
| Risk factors begin accumulating | 224 | 93 | 41.5% |
| Moderate risk period | 382 | 223 | 58.4% |
| Higher risk period | 197 | 145 | 73.6% |
| Highest risk population | 24 | 16 | 66.7% |

# Clinical Focus – Blood Pressure

**Blood Pressure Categories in our Data**

- 161 Normal
- 211 Elevated
- 177 Stage 1 HTN
- 311 Stage 2 HTN

**Percentage with Heart Disease**

- Normal 54.04%
- Elevated 49.29%
- Stage 1 HTN 48.59%
- Stage 2 HTN 61.74%

**Blood Pressure Categories**

American Heart Association | American Stroke Association

| BLOOD PRESSURE CATEGORY | SYSTOLIC mm Hg (upper number) | | DIASTOLIC mm Hg (lower number) |
|---|---|---|---|
| NORMAL | LESS THAN 120 | and | LESS THAN 80 |
| ELEVATED | 120 – 129 | and | LESS THAN 80 |
| HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1 | 130 – 139 | or | 80 – 89 |
| HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2 | 140 OR HIGHER | or | 90 OR HIGHER |
| HYPERTENSIVE CRISIS (consult your doctor immediately) | HIGHER THAN 180 | and/or | HIGHER THAN 120 |

American Heart Association Classifiers

Blood pressure is very loosely correlated to heart disease, with a U shaped distribution starting high, going lower and then peaking. However due to the prevalence of medication, many of those in the Normal category may be there due to medication prescribed for their heart disease, lowering their blood pressure.

# Clinical Focus - Cholesterol Risk

| Cholesterol Category | Range(mg/dL) | Count | Rate(Exclusive Missing)(%) |
| --- | --- | --- | --- |
| Missing/Invalid | 0 or NaN | 202 | |
| Desirable | < 200 | 128 | 37.50 |
| Borderline High | 200-239 | 231 | 46.32 |
| High | ≥ 240 | 359 | 53.20 |

As cholesterol category increases, heart disease prevalence increases.

# Clinical Focus - Heart Rate response

```
# Step 1: Calculate age-predicted maximum heart rate
df["max_hr_predicted"] = 220 - df["age"]


# Step 2: Calculate percentage of maximum achieved
df["percent_max_hr"] = (df["thalch"] / df["max_hr_predicted"]) * 100


# Step 3: Create the hr_response category column
def classify_hr(row):
    if pd.isna(row["thalch"]) or row["thalch"] == 0:
        return "Missing/Invalid"
    elif row["percent_max_hr"] < 70:
        return "Poor Response"
    elif row["percent_max_hr"] < 85:
        return "Submaximal"
    elif row["percent_max_hr"] < 100:
        return "Adequate"
    else:
        return "Excellent"

df["hr_response"] = df.apply(classify_hr, axis=1)
```

```
### Verify your groupings worked:
df["hr_response"].value_counts()

# Show counts per heart rate response category
```

```
hr_response
Adequate           292
Submaximal         290
Poor Response      179
Excellent          104
Missing/Invalid     55
Name: count, dtype: int64
```

```
### Calculate heart disease rate by heart rate response:
df["disease_binary"] = (df["num"] > 0).astype(int)

(
    df[df["hr_response"] != "Missing/Invalid"]
    .groupby("hr_response")["disease_binary"]
    .mean()
    .mul(100)
    .round(2)
)


# Create a binary indicator for heart disease (num > 0 means disease present)
# Then calculate the percentage of patients with heart disease in each HR response category
# Exclude the "Missing/Invalid" category from your interpretation
```

```
hr_response
Adequate         43.49
Excellent        31.73
Poor Response    75.98
Submaximal       61.38
Name: disease_binary, dtype: float64
```

- Calculated predicted max HR (220 − age)
- Computed % of predicted HR achieved ((thalch / predicted) × 100)
- Categorized response: Poor, Submaximal, Adequate, Excellent
- Poor Response → **75.98%** heart disease
- Submaximal → **61.38%** heart disease
- Adequate → **43.49%** heart disease
- Excellent → **31.73%** heart disease

Lower HR response → Higher heart disease rate

# How do these tie together?

- We can see that the risk of heart disease is tied to conventional factors
  - Old age, high cholesterol levels, abnormal blood pressure, and low heart response are all factors commonly found in patients with some level of heart disease.
- All these factors can be used as reliable predictors for training a model to predict heart disease

# Recommendations

For data providers

- Improving data collection methods
  - Adding new measurements such as medication status
  - Standardize what is collected

For Clinics

- Prioritize patients with Poor Heart Rate Response (~76% suffer from disease)
- Prioritize patients with Stage 2 Hypertension (62% risk)

For Consulting Firm

- Develop a machine learning model to predict patient outcomes
  - Begin with a binary classifier and work from there to improve to predict more complex outcomes